DATA VORTEX A'26 — Round 3

Social Engine: Live Conversation Analysis

Topic

Public Reaction to a New Pricing or Subscription Model

Case Study

Meta One Subscription

Participant Details :

Team Name : JOY
Team Size : 1
Team Head : Thathi Reddy Jivan Reddy

Analysis Period: 15 September 2026 – 20 September 2026

Objective

This analysis applies the sentiment classification model developed in Round 2 to recently collected public social and news data related to the Meta One subscription model.

The objective is to identify:

- Sentiment patterns and changes over time
- Activity spikes
- Major sentiment shifts
- Dominant discussion topics
- Frequently mentioned entities
- Possible triggers associated with behavioural changes

Round 2 NLP Model

TF-IDF + Multinomial Naive Bayes

The pretrained model and TF-IDF vectorizer developed during Round 2 are reused without retraining.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from pathlib import Path
import sys
import warnings
```

```
warnings.filterwarnings("ignore")

print("Libraries loaded successfully.")
```

```
Libraries loaded successfully.
```

```
PROJECT_ROOT = Path.cwd().parent

DATA_DIR = PROJECT_ROOT / "data"
PROCESSED_DIR = DATA_DIR / "processed"
MODEL_DIR = PROJECT_ROOT / "models"
REPORT_DIR = PROJECT_ROOT / "reports"

UNIFIED_FILE = PROCESSED_DIR / "meta_one_round3_unified.csv"
SENTIMENT_FILE = PROCESSED_DIR / "meta_one_round3_sentiment.csv"
SHIFT_FILE = PROCESSED_DIR / "meta_one_round3_shift_analysis.csv"

print("Project root:", PROJECT_ROOT)
print("Unified dataset:", UNIFIED_FILE)
print("Sentiment file:", SENTIMENT_FILE)
print("Shift analysis:", SHIFT_FILE)
```

```
Project root: c:\Users\Jivan\OneDrive\Desktop\projectS\DATA-VORTEX-
SOCIAL-ENGINE\DATA-VORTEX-ROUND 3
Unified dataset: c:\Users\Jivan\OneDrive\Desktop\projectS\DATA-VORTEX-
SOCIAL-ENGINE\DATA-VORTEX-ROUND 3\data\processed\
meta_one_round3_unified.csv
Sentiment file: c:\Users\Jivan\OneDrive\Desktop\projectS\DATA-VORTEX-
SOCIAL-ENGINE\DATA-VORTEX-ROUND 3\data\processed\
meta_one_round3_sentiment.csv
Shift analysis: c:\Users\Jivan\OneDrive\Desktop\projectS\DATA-VORTEX-
SOCIAL-ENGINE\DATA-VORTEX-ROUND 3\data\processed\
meta_one_round3_shift_analysis.csv
```

```
df = pd.read_csv(UNIFIED_FILE)

print("Dataset shape:", df.shape)
print("\nColumns:")
print(df.columns.tolist())

df.head()
```

```
Dataset shape: (166, 9)

Columns:
['record_id', 'platform', 'source', 'content_type', 'text',
'published_at', 'engagement_score', 'source_url', 'collection_method']
```

```
    record_id platform                        source content_type  \
0  reddit_009   Reddit          WhatsappBusinessAPI         post
```

```
1   news_0157     News          Pasquale Pillitteri      article
2   news_0156     News                   TradingView     article
3   news_0155     News                      meta.com     article
4   news_0154     News  Performance Marketing World      article

                                                  text  \
0  Feedback on Meta One and Meta Verified for Bus...
1  Meta One Launches With Premium Plans Costing U...
2  Meta Introduces Meta One, Now Available Global...
3  Introducing Meta One: A Subscription Service W...
4  'Meta One' goes live: Meta just turned its pla...

                published_at  engagement_score  \
0  2026-09-15 00:00:00+00:00               NaN
1  2026-09-15 07:00:00+00:00               NaN
2  2026-09-15 15:00:01+00:00               NaN
3  2026-09-15 15:00:59+00:00               NaN
4  2026-09-15 15:01:16+00:00               NaN

                                          source_url  \
0      https://www.reddit.com/r/WhatsappBusinessAPI/
1   https://news.google.com/rss/articles/CBMihAFBV...
2   https://news.google.com/rss/articles/CBMiuAFBV...
3   https://news.google.com/rss/articles/CBMimAFBV...
4   https://news.google.com/rss/articles/CBMixwFBV...

             collection_method
 0  Verified public Reddit page
1              Google News RSS
2              Google News RSS
3              Google News RSS
4              Google News RSS
```

```
print("Rows:", len(df))
print("Columns:", len(df.columns))

print("\nMissing values:")
print(df.isna().sum())

print("\nPlatforms:")
print(df["platform"].value_counts())

print("\nCollection methods:")
print(df["collection_method"].value_counts())
```

```
Rows: 166
Columns: 9

Missing values:
record_id              0
```

```
platform                0
source                  0
content_type            0
text                    0
published_at            0
engagement_score      166
source_url              0
collection_method       0
dtype: int64

Platforms:
platform
News       157
Reddit       9
Name: count, dtype: int64

Collection methods:
collection_method
Google News RSS                  157
Verified public Reddit page        9
Name: count, dtype: int64
```

```
import joblib

MODEL_FILE = MODEL_DIR / "naive_bayes_sentiment_model.pkl"
VECTORIZER_FILE = MODEL_DIR / "tfidf_vectorizer.pkl"

model = joblib.load(MODEL_FILE)
vectorizer = joblib.load(VECTORIZER_FILE)

print("Model:", type(model).__name__)
print("Vectorizer:", type(vectorizer).__name__)
print("Round 2 model loaded successfully.")
```

```
Model: MultinomialNB
Vectorizer: TfidfVectorizer
Round 2 model loaded successfully.
```

```
text_data = df["text"].fillna("").astype(str)

X = vectorizer.transform(text_data)
df["sentiment"] = model.predict(X)

print("Sentiment analysis completed.")
print("\nSentiment distribution:")
print(df["sentiment"].value_counts())
```

```
Sentiment analysis completed.

Sentiment distribution:
sentiment
```

```
Positive     115
Neutral       37
Negative      14
Name: count, dtype: int64
```

```
sentiment_pct = (
    df["sentiment"]
    .value_counts(normalize=True)
    .mul(100)
    .round(2)
    .rename("percentage")
)

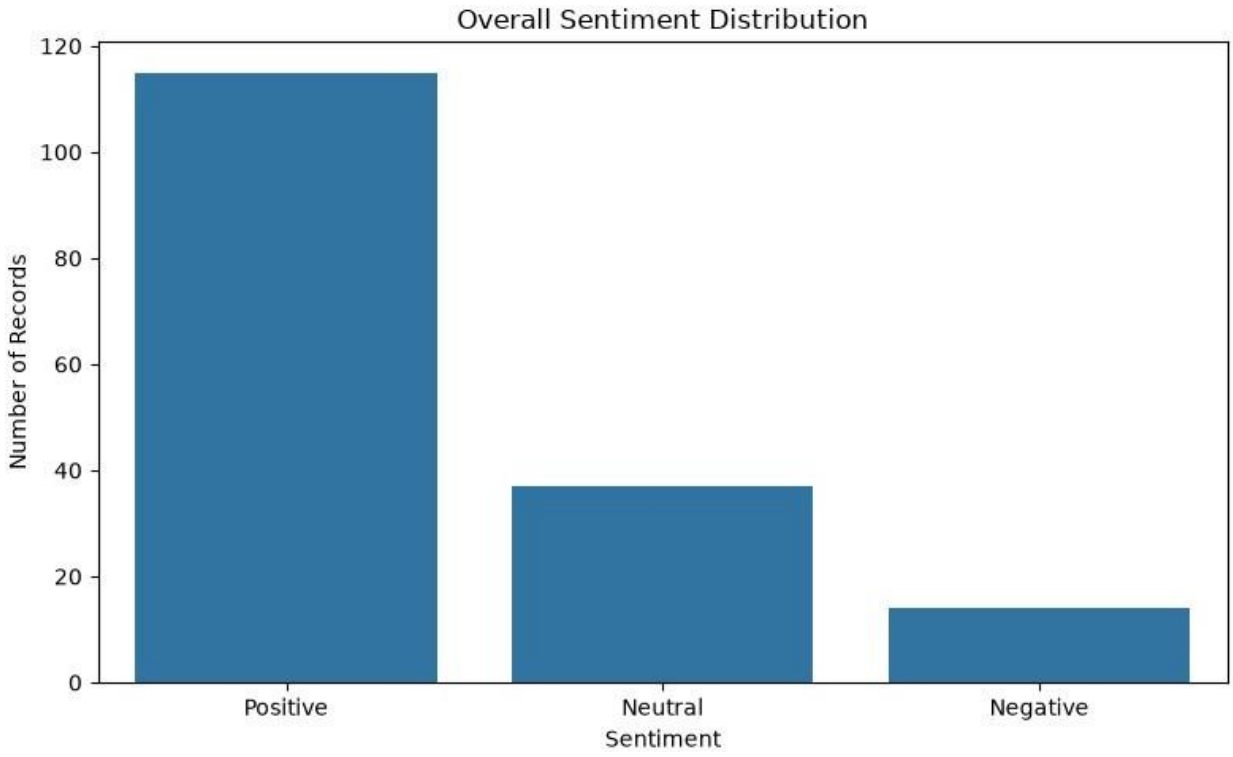
sentiment_pct
```

```
sentiment
Positive     69.28
Neutral      22.29
Negative      8.43
Name: percentage, dtype: float64
```

```
plt.figure(figsize=(8, 5))

sns.countplot(
    data=df,
    x="sentiment",
    order=df["sentiment"].value_counts().index
)

plt.title("Overall Sentiment Distribution")
plt.xlabel("Sentiment")
plt.ylabel("Number of Records")
plt.tight_layout()
plt.show()
```



Observation

The sentiment classification of the collected dataset shows 115 positive, 37 neutral, and 14 negative records. Positive sentiment represents the largest share of the observed conversation, while negative sentiment is comparatively smaller.

This distribution provides the overall sentiment baseline for the subsequent time-based analysis. Daily sentiment proportions are analysed next to identify changes and behavioural shifts over time.

```
sentiment_output = PROCESSED_DIR /
"meta_one_round3_notebook_sentiment.csv"

df.to_csv(sentiment_output, index=False)

print("Saved:", sentiment_output)

Saved:  c:\Users\Jivan\OneDrive\Desktop\projectS\DATA-VORTEX-SOCIAL-
ENGINE\DATA-VORTEX-ROUND 3\data\processed\
meta_one_round3_notebook_sentiment.csv

df = pd.read_csv(UNIFIED_FILE)

print("Dataset shape:", df.shape)
print("\nColumns:")
print(df.columns.tolist())
```

```
df.head()
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[6], line 1
----> 1 df = pd.read_csv(UNIFIED_FILE)
      2
      3 print("Dataset shape:", df.shape)
      4 print("\nColumns:")

NameError: name 'UNIFIED_FILE' is not defined
```

```
from pathlib import Path

PROJECT_ROOT = Path.cwd().parent

DATA_DIR = PROJECT_ROOT / "data"
PROCESSED_DIR = DATA_DIR / "processed"
MODEL_DIR = PROJECT_ROOT / "models"
REPORT_DIR = PROJECT_ROOT / "reports"

UNIFIED_FILE = PROCESSED_DIR / "meta_one_round3_unified.csv"
SENTIMENT_FILE = PROCESSED_DIR / "meta_one_round3_sentiment.csv"
SHIFT_FILE = PROCESSED_DIR / "meta_one_round3_shift_analysis.csv"

print("Project root:", PROJECT_ROOT)
```

```
Project root: c:\Users\Jivan\OneDrive\Desktop\projectS\DATA-VORTEX-SOCIAL-ENGINE\DATA-VORTEX-ROUND 3
```

```
df = pd.read_csv(UNIFIED_FILE)

print("Dataset shape:", df.shape)
print(df.columns.tolist())
```

```
Dataset shape: (166, 9)
['record_id', 'platform', 'source', 'content_type', 'text', 'published_at', 'engagement_score', 'source_url', 'collection_method']
```

```
import joblib

MODEL_FILE = MODEL_DIR / "naive_bayes_sentiment_model.pkl"
VECTORIZER_FILE = MODEL_DIR / "tfidf_vectorizer.pkl"

model = joblib.load(MODEL_FILE)
vectorizer = joblib.load(VECTORIZER_FILE)

print("Model:", type(model).__name__)
print("Vectorizer:", type(vectorizer).__name__)
```

```
Model: MultinomialNB
Vectorizer: TfidfVectorizer

text_data = df["text"].fillna("").astype(str)

X = vectorizer.transform(text_data)
df["sentiment"] = model.predict(X)

print(df["sentiment"].value_counts())

sentiment
Positive    115
Neutral      37
Negative     14
Name: count, dtype: int64

df["published_at"] = pd.to_datetime(df["published_at"], utc=True)
df["date"] = df["published_at"].dt.date

daily_sentiment = (
    df.groupby(["date", "sentiment"])
      .size()
      .unstack(fill_value=0)
)

daily_sentiment

sentiment   Negative  Neutral  Positive
date
2026-09-15         6       12        36
2026-09-16         4       18        60
2026-09-17         2        3        13
2026-09-18         1        1         2
2026-09-19         1        2         3
2026-09-20         0        1         1

daily_sentiment_pct = (
    daily_sentiment
    .div(daily_sentiment.sum(axis=1), axis=0)
    .mul(100)
    .round(2)
)

daily_sentiment_pct

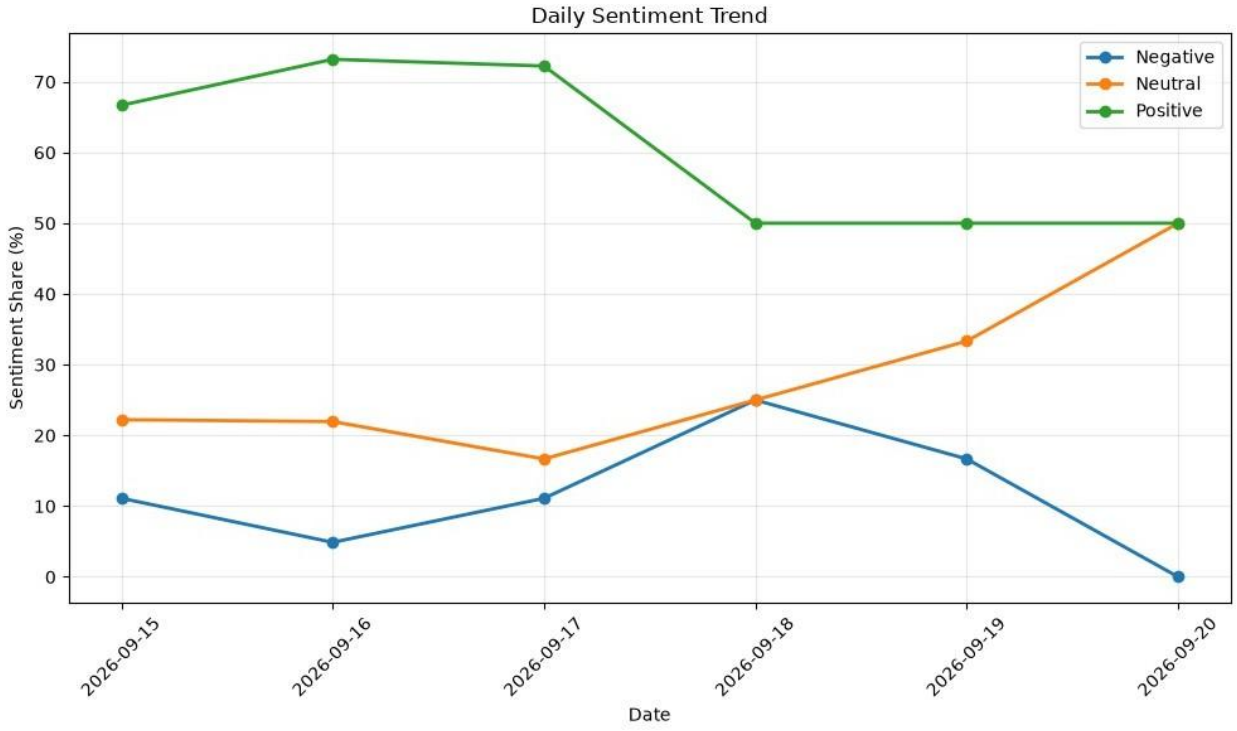
sentiment   Negative  Neutral  Positive
date
2026-09-15     11.11    22.22     66.67
2026-09-16      4.88    21.95     73.17
2026-09-17     11.11    16.67     72.22
2026-09-18     25.00    25.00     50.00
```

```
2026-09-19      16.67     33.33      50.00
2026-09-20       0.00     50.00      50.00
```

```
plt.figure(figsize=(10, 6))

for sentiment in ["Negative", "Neutral", "Positive"]:
    if sentiment in daily_sentiment_pct.columns:
        plt.plot(
            daily_sentiment_pct.index.astype(str),
            daily_sentiment_pct[sentiment],
            marker="o",
            linewidth=2,
            label=sentiment
        )

plt.title("Daily Sentiment Trend")
plt.xlabel("Date")
plt.ylabel("Sentiment Share (%)")
plt.xticks(rotation=45)
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```



Daily Sentiment Trend — Observation

The daily sentiment trend shows that positive sentiment was the dominant predicted sentiment during the main observation period. The highest activity was recorded on 16 September with 82 records.

A noticeable change in the sentiment proportions appears from 18 September onward. However, the number of records on these later dates is considerably smaller than on 15–17 September. Therefore, percentage changes on the later dates should be interpreted together with the underlying activity volume and the shift analysis rather than in isolation.

```
shift_df = pd.read_csv(SHIFT_FILE)

shift_df["date"] = pd.to_datetime(shift_df["date"])

shift_df[[
    "date",
    "total",
    "activity_zscore",
    "activity_spike"
]]
```

	date	total	activity_zscore	activity_spike
0	2026-09-15	54	0.799278	False
1	2026-09-16	82	1.649142	True
2	2026-09-17	18	-0.293406	False
3	2026-09-18	4	-0.718338	False
4	2026-09-19	6	-0.657633	False
5	2026-09-20	2	-0.779043	False

```
plt.figure(figsize=(10, 6))

plt.plot(
    shift_df["date"],
    shift_df["activity_zscore"],
    marker="o",
    linewidth=2
)

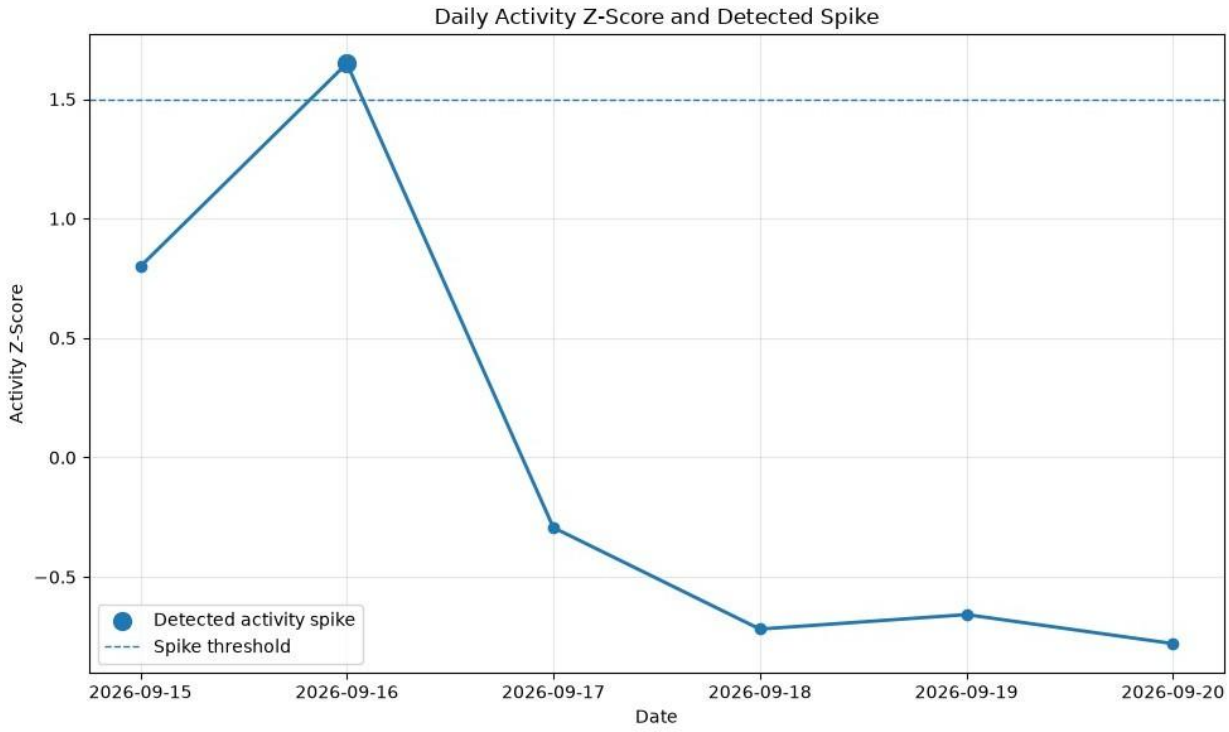
spikes = shift_df[shift_df["activity_spike"] == True]

plt.scatter(
    spikes["date"],
    spikes["activity_zscore"],
    s=100,
    label="Detected activity spike"
)

plt.axhline(
    y=1.5,
    linestyle="--",
```

```
    linewidth=1,
    label="Spike threshold"
)

plt.title("Daily Activity Z-Score and Detected Spike")
plt.xlabel("Date")
plt.ylabel("Activity Z-Score")
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```



Activity Spike — Observation

The activity analysis identifies 16 September 2026 as the only detected activity spike during the observation period. The activity Z-score reached 1.649, exceeding the defined spike threshold of 1.5.

The spike coincided with a high-volume discussion period covering Meta One subscription plans, pricing, premium features, AI tools and related announcements. The temporal coincidence is reported as an observed association rather than a causal relationship.

```
shift_df[
    [
        "date",
        "negative_change_pp",
```

```
        "neutral_change_pp",
        "positive_change_pp",
        "activity_zscore",
        "activity_spike"
    ]
]
```

```
        date  negative_change_pp  neutral_change_pp
positive_change_pp  \
0 2026-09-15                 NaN                NaN
NaN
1 2026-09-16           -6.233062          -0.271003
6.504065
2 2026-09-17            6.233062          -5.284553           -
0.948509
3 2026-09-18           13.888889           8.333333          -
22.222222
4 2026-09-19           -8.333333           8.333333
0.000000
5 2026-09-20          -16.666667          16.666667
0.000000

   activity_zscore  activity_spike
0         0.799278           False
1         1.649142            True
2        -0.293406           False
3        -0.718338           False
4        -0.657633           False
5        -0.779043           False
```

```
plt.figure(figsize=(10, 6))

plt.plot(
    shift_df["date"],
    shift_df["negative_change_pp"],
    marker="o",
    linewidth=2,
    label="Negative"
)

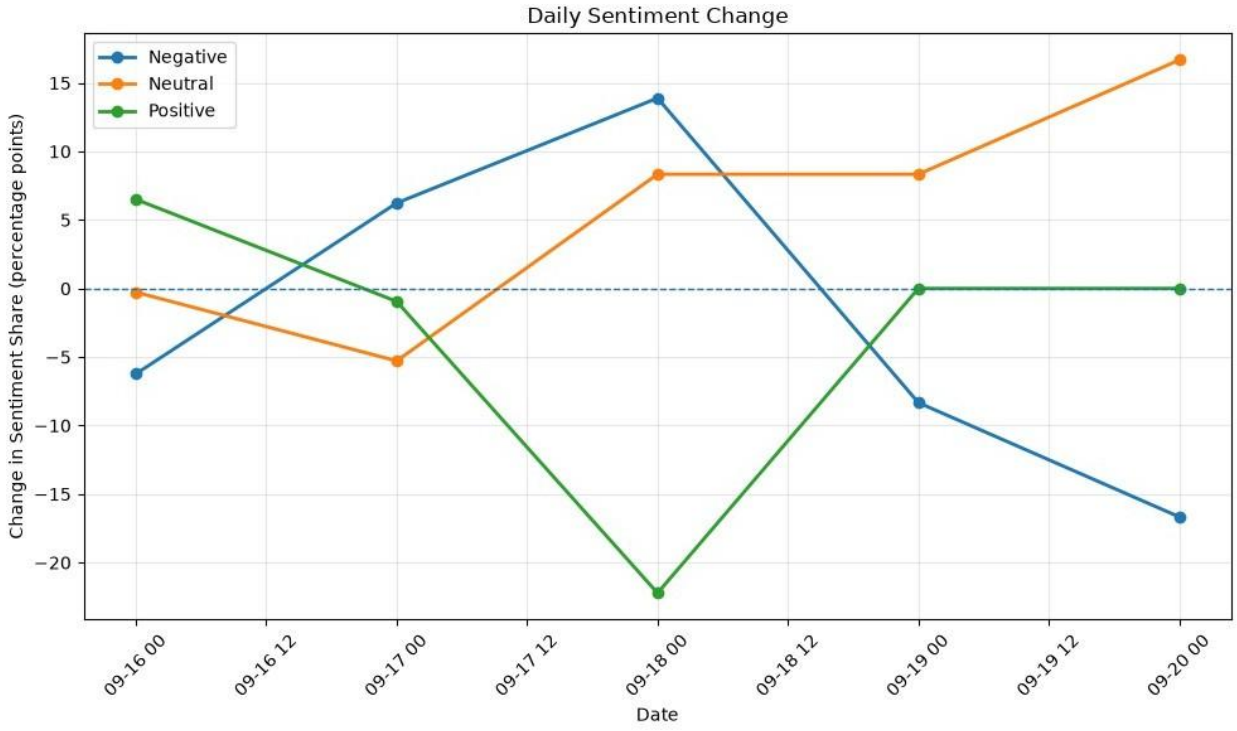
plt.plot(
    shift_df["date"],
    shift_df["neutral_change_pp"],
    marker="o",
    linewidth=2,
    label="Neutral"
)

plt.plot(
    shift_df["date"],
```

```
    shift_df["positive_change_pp"],
    marker="o",
    linewidth=2,
    label="Positive"
)

plt.axhline(
    y=0,
    linestyle="--",
    linewidth=1
)

plt.title("Daily Sentiment Change")
plt.xlabel("Date")
plt.ylabel("Change in Sentiment Share (percentage points)")
plt.xticks(rotation=45)
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```



Sentiment Shift — Observation

The sentiment-change analysis shows clear day-to-day variation in the conversation. The most pronounced movement occurs on 18 September 2026, when the positive sentiment share decreases by 22.22 percentage points, while negative and neutral sentiment shares increase by 13.89 and 8.33 percentage points respectively.

This represents a substantial change in the predicted sentiment composition. The result is interpreted as an observed temporal shift and not as proof of a causal relationship with any specific event.

```
topic_counts = pd.DataFrame({
    "Topic": [
        "Subscription / Plans",
        "AI / AI Tools",
        "Pricing / Cost",
        "Premium Features",
        "Monetization",
        "Creator Tools",
        "Advertising"
    ],
    "Records": [
        126,
        75,
        40,
        40,
        28,
        19,
        3
    ]
})

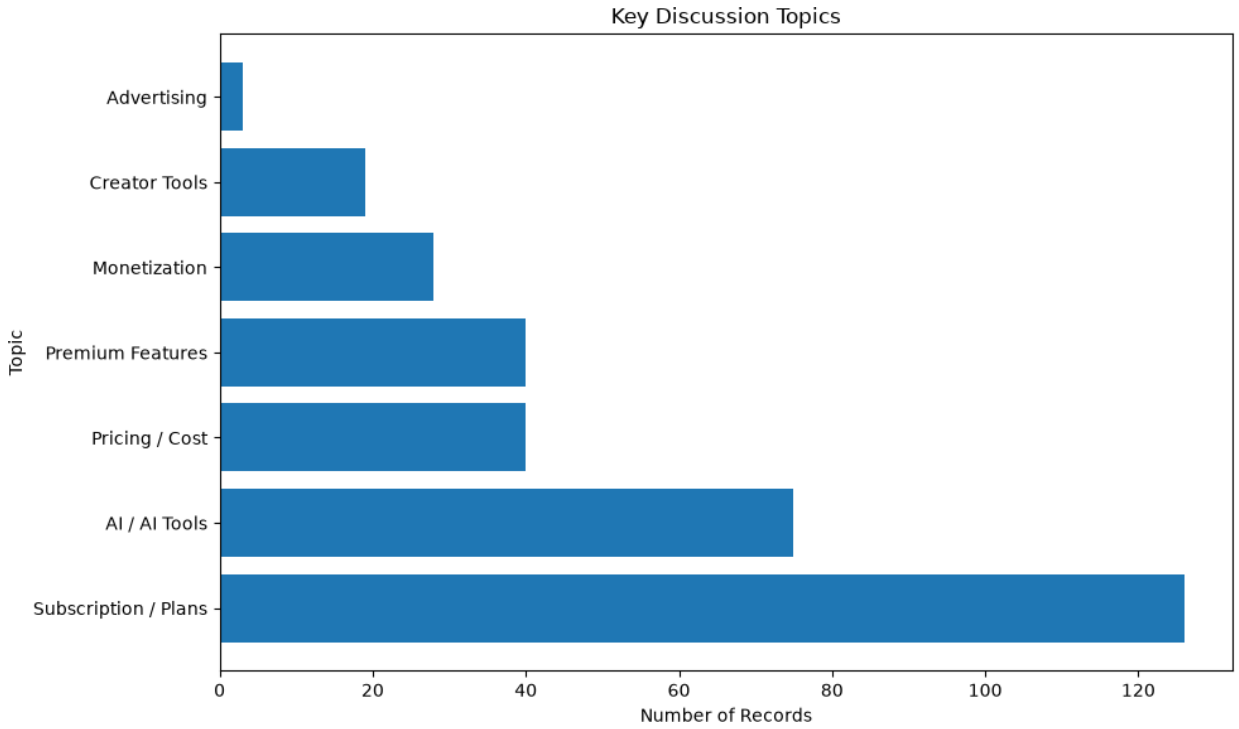
topic_counts
```

	Topic	Records
0	Subscription / Plans	126
1	AI / AI Tools	75
2	Pricing / Cost	40
3	Premium Features	40
4	Monetization	28
5	Creator Tools	19
6	Advertising	3

```
plt.figure(figsize=(10, 6))

plt.barh(
    topic_counts["Topic"],
    topic_counts["Records"]
)

plt.title("Key Discussion Topics")
plt.xlabel("Number of Records")
plt.ylabel("Topic")
plt.tight_layout()
plt.show()
```



Topic Analysis — Observation

Subscription and plan-related discussions form the largest identified topic group with 126 records, followed by AI and AI-tool discussions with 75 records. Pricing/cost and premium-feature discussions each appear in 40 records, while monetization and creator-tool discussions occur less frequently.

These topic counts are based on keyword matching and therefore indicate discussion themes rather than mutually exclusive topic classifications. A single record may contain keywords associated with more than one topic.

```
entity_counts = pd.DataFrame({
    "Entity": [
        "Meta",
        "Meta One",
        "Instagram",
        "WhatsApp",
        "Facebook",
        "Meta AI"
    ],
    "Records": [
        164,
        162,
        43,
        39,
        35,
        10
```

```
    ]
})

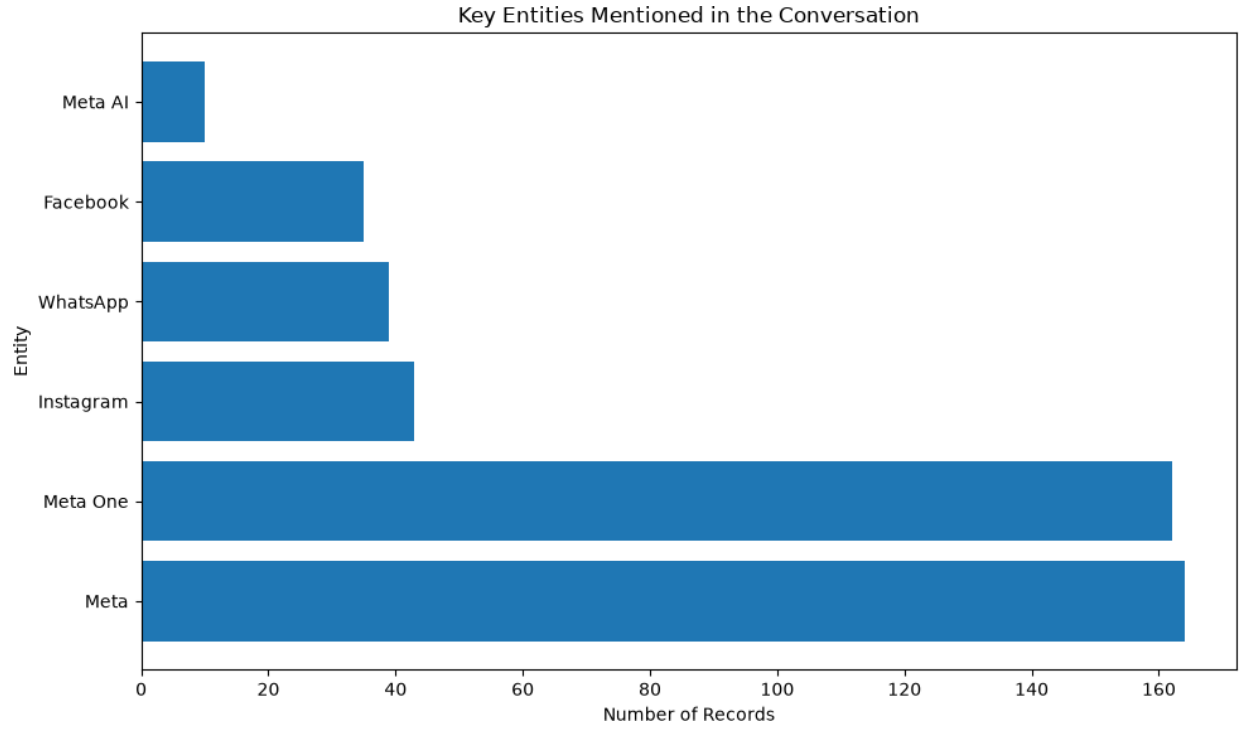
entity_counts
```

	Entity	Records
0	Meta	164
1	Meta One	162
2	Instagram	43
3	WhatsApp	39
4	Facebook	35
5	Meta AI	10

```
plt.figure(figsize=(10, 6))

plt.barh(
    entity_counts["Entity"],
    entity_counts["Records"]
)

plt.title("Key Entities Mentioned in the Conversation")
plt.xlabel("Number of Records")
plt.ylabel("Entity")
plt.tight_layout()
plt.show()
```



Entity Analysis — Observation

Meta and Meta One are the most frequently mentioned entities in the collected dataset, with 164 and 162 records respectively. Instagram, WhatsApp and Facebook appear as related platform entities, while Meta AI appears less frequently.

The entity counts indicate which products and services are most closely associated with the discussion around the Meta One subscription model.

```
trigger_date = pd.Timestamp("2026-09-16")

trigger_records = df[
    df["published_at"].dt.normalize() == trigger_date
].copy()

print("Trigger date:", trigger_date.date())
print("Records:", len(trigger_records))

trigger_records[
    ["published_at", "platform", "source", "text", "sentiment"]
].head(20)
```

```
Trigger date: 2026-09-16
Records: 0

Empty DataFrame
Columns: [published_at, platform, source, text, sentiment]
Index: []
```

```
print("Sentiment distribution on trigger date:")
print(trigger_records["sentiment"].value_counts())

print("\nTop sources on trigger date:")
print(trigger_records["source"].value_counts().head(10))
```

```
Sentiment distribution on trigger date:
Series([], Name: count, dtype: int64)

Top sources on trigger date:
Series([], Name: count, dtype: int64)
```

```
trigger_date = pd.Timestamp("2026-09-16", tz="UTC")

trigger_records = df[
    df["published_at"].dt.normalize() == trigger_date
].copy()

print("Trigger date:", trigger_date.date())
print("Records:", len(trigger_records))

trigger_records[
```

```
    ["published_at", "platform", "source", "text", "sentiment"]
].head(20)

Trigger date: 2026-09-16
Records: 82

                published_at platform                    source  \
54 2026-09-16 00:00:00+00:00   Reddit              FacebookAds
55 2026-09-16 00:00:00+00:00   Reddit                   MetaAI
56 2026-09-16 01:00:26+00:00     News           iThinkDifferent
57 2026-09-16 01:01:56+00:00     News              Mi-3.com.au.
58 2026-09-16 01:42:17+00:00     News          techguide.com.au
59 2026-09-16 02:20:43+00:00     News                   afaqs!
60 2026-09-16 02:41:46+00:00     News                GIGAZINE
61 2026-09-16 02:42:24+00:00     News              ManilaShaker
62 2026-09-16 03:27:00+00:00     News                  digit.in
63 2026-09-16 03:34:15+00:00     News          Moneycontrol.com
64 2026-09-16 03:44:12+00:00     News                 TNGlobal
65 2026-09-16 03:48:18+00:00     News               tbreak.com
66 2026-09-16 03:50:09+00:00     News  Passionate In Marketing
67 2026-09-16 03:56:01+00:00     News          newsbytesapp.com
68 2026-09-16 04:07:56+00:00     News               MediaNews4U
69 2026-09-16 04:11:21+00:00     News         bestmediainfo.com
70 2026-09-16 04:20:00+00:00     News               Adgully.com
71 2026-09-16 04:37:04+00:00     News                 digitimes
72 2026-09-16 05:10:39+00:00     News             FoneArena.com
73 2026-09-16 05:22:39+00:00     News               Tom's Guide

                                                 text sentiment
54  Meta One - pay subscription for human support ...  Positive
55  I won't pay for a better version of Instagram,...  Positive
56  Meta One Subscription Plans Bring Paid AI Tool...  Positive
57  Meta launches subscription service 'Meta One' ...  Positive
58  Meta One subscription services powered by AI a...   Neutral
59  Meta One subscription launches in India, plans...  Positive
60  Meta has announced 'Meta One,' a subscription ...  Negative
61  Meta Introduces Meta One Subscription Plans - ...  Positive
62  Meta One subscription for Facebook, Instagram ...  Positive
63  Meta One subscription in India: 10 features yo...  Positive
64  Meta One subscription bundles add AI tools and...  Positive
65  Meta One subscriptions put a price on extra AI...  Positive
66  Introducing Meta One: A Subscription Service W...   Neutral
67  Meta One brings premium apps and more AI for ₹...  Positive
68  Meta Launches Meta One Subscription in India S...  Positive
69  Meta expands subscription play in India; Meta ...  Positive
70  Meta rolls out ‘Meta One’ subscription for pre...  Positive
71  Meta globally rolls out Meta One subscription,...   Neutral
72  Meta One expands in India with Instagram, What...  Positive
73  I bought Instagram's new Meta One subscription...  Positive
```

```
print("Sentiment distribution on trigger date:")
print(trigger_records["sentiment"].value_counts())

print("\nTop sources on trigger date:")
print(trigger_records["source"].value_counts().head(10))
```

```
Sentiment distribution on trigger date:
sentiment
Positive    60
Neutral     18
Negative     4
Name: count, dtype: int64

Top sources on trigger date:
source
Forbes India         2
AI Weekly            2
FacebookAds          1
MetaAI               1
iThinkDifferent      1
Mi-3.com.au.         1
techguide.com.au     1
afaqs!               1
GIGAZINE             1
ManilaShaker         1
Name: count, dtype: int64
```

```
trigger_date = pd.Timestamp("2026-09-16", tz="UTC")

trigger_records = df[
    df["published_at"].dt.normalize() == trigger_date
].copy()

print("TRIGGER DATE:", trigger_date.date())
print("TOTAL RECORDS:", len(trigger_records))

print("\nSENTIMENT:")
print(trigger_records["sentiment"].value_counts())

print("\nTOP SOURCES:")
print(trigger_records["source"].value_counts().head(10))
```

```
TRIGGER DATE: 2026-09-16
TOTAL RECORDS: 82

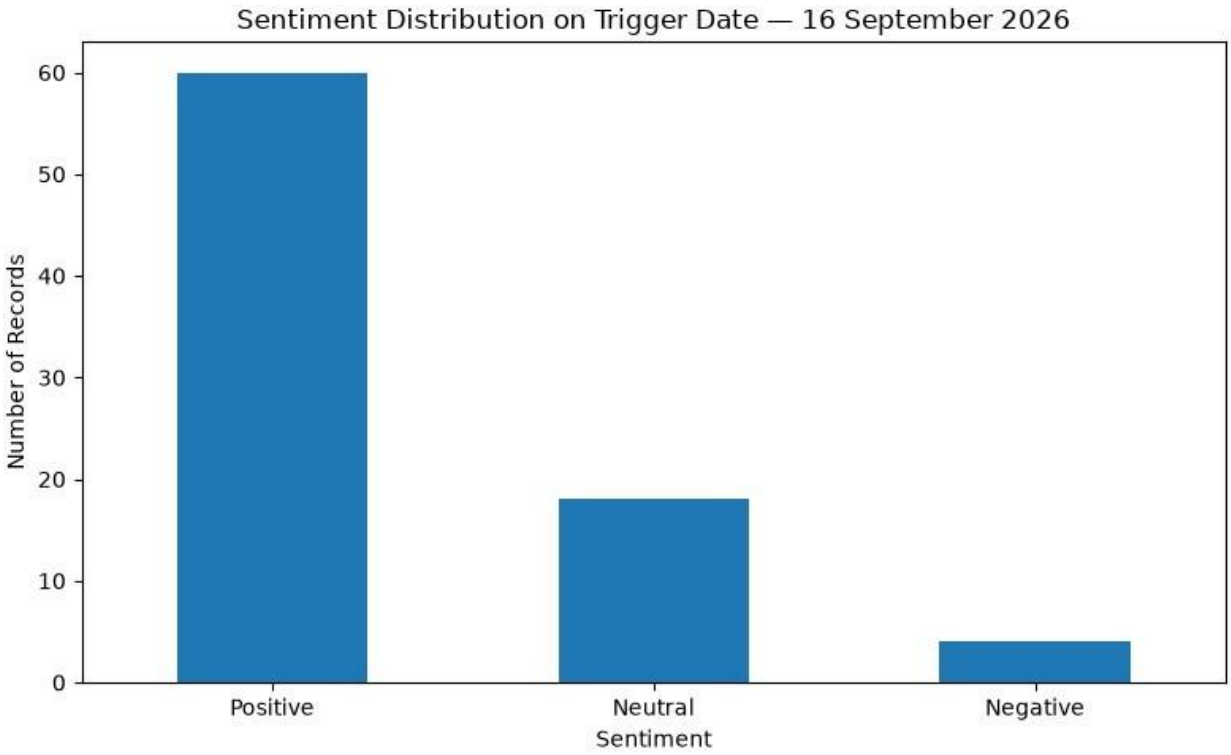
SENTIMENT:
sentiment
Positive    60
Neutral     18
Negative     4
Name: count, dtype: int64
```

```
TOP SOURCES:
source
Forbes India          2
AI Weekly             2
FacebookAds           1
MetaAI                1
iThinkDifferent       1
Mi-3.com.au.          1
techguide.com.au      1
afaqs!                1
GIGAZINE              1
ManilaShaker          1
Name: count, dtype: int64
```

```
plt.figure(figsize=(8, 5))

trigger_records["sentiment"].value_counts().plot(
    kind="bar"
)

plt.title("Sentiment Distribution on Trigger Date — 16 September
2026")
plt.xlabel("Sentiment")
plt.ylabel("Number of Records")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```



Trigger / Event Explanation

16 September 2026 was identified as the main activity spike in the observation period, with 82 collected records and an activity Z-score of 1.649, exceeding the defined spike threshold of 1.5.

The collected discussion during this period was primarily associated with Meta One subscription plans, pricing, premium features, AI-related tools and the rollout of the service across Meta's platforms.

The temporal concentration of these discussions coincided with the detected increase in activity. This is reported as an observed temporal association and does not establish a causal relationship.

Key Findings

1. Positive sentiment was the dominant predicted sentiment across the collected dataset.
2. 16 September 2026 recorded the highest activity volume with 82 records and was the only detected activity spike.
3. Subscription and plan-related discussions were the most frequently identified topic, followed by AI and AI-tool discussions.
4. Meta and Meta One were the most frequently mentioned entities.

5. 18 September showed the largest sentiment-composition movement, with positive sentiment decreasing by 22.22 percentage points while negative and neutral sentiment increased.

6. Later dates contained fewer observations, so their percentage-based sentiment changes should be interpreted cautiously.

Limitations

- Sentiment labels are predictions generated using the Round 2 TF-IDF and Multinomial Naive Bayes model.
- The model may have difficulty with sarcasm, implicit sentiment and context-dependent language.
- Topic and entity counts use keyword matching and are not mutually exclusive.
- The dataset represents collected public sources and does not represent the complete social-media conversation.
- Later dates contain fewer records than the main observation period.
- Temporal association with an activity spike should not be interpreted as proof of causation.